

# MECHATRONICS ENGINEER

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## SUMMARY

Mechatronics and mechanical design engineer with hands-on experience across product design and industrial/mechatronic systems, from smart-home consumer devices to steel-production line equipment. Comfortable owning a design end-to-end — concept, CAD, prototyping, and production handoff — and collaborating directly with founders and cross-functional teams (production, electrical, automation).

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## WORK EXPERIENCE

### Mechatronics Engineer, R&D — Supishi, Tehran, Iran

Jun 2026 – Present

- Design production-line equipment, material handling systems, and industrial conveyors within the R&D unit.
- Produce 3D designs and technical drawings, and select and calculate mechanical components for industrial equipment.
- Collaborate with production, electrical, and automation teams to deliver industrial projects end-to-end.

### Product Designer — NYOP, Amsterdam, Netherlands (Remote)

Aug 2025 – Present

- Design and develop smart-home products, including a smart insect trap and an air-quality monitoring device (temperature, humidity, dust, pH).
- Work directly with the CEO across ideation, concept design, prototyping, and production readiness.
- Use CAD software for mechanical design and iterate on designs based on test data and user feedback.

### Mechanical Designer & Engineer — Iranian Steel Development Group, Tabriz, Iran

May 2023 – May 2024

- Led design and development of production-line equipment for a steel manufacturing plant, including mechanical design of industrial machine components.
- Oversaw part manufacturing and contributed to production-process optimization and production scheduling.

### SolidWorks Industrial Designer — Mechatronics Mobtakeran Arg, Tabriz, Iran

Apr 2022 – Sep 2022

- Designed components for laboratory 3D printers, dental milling machines, intraoral and laboratory scanners, and casting-resin systems.
- Modeled parts in SolidWorks and ExoCAD for dental/medical device manufacturing.

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## SELECTED PROJECTS

- Tuned reinforcement-learning reward functions using large language models for mobile robot navigation (2025)
- Modeled vibration and optimized a bladeless vortex-induced wind turbine with a tuning system to increase efficiency (2023)
- Designed, assembled, and animated a housingless stand for a steel production line (2023)
- Designed a VIVACE energy-harvesting device for low-speed wave environments (2022)
- Designed a water channel with uniform flow to study vibration of an oscillating cylinder (2022)
- Simulated lift and drag forces on a circular cylinder in laminar flow (2021)

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## LANGUAGES & TECHNOLOGIES

**CAD & Simulation:** SolidWorks, CATIA, AutoCAD, Ansys Fluent, KISSsoft

**Automation & Systems:** Siemens LOGO! PLC programming, hydraulic & pneumatic systems (FluidSIM), Arduino

**Programming & Analysis:** Python, MATLAB, data structures, deep learning

**Project Tools:** Microsoft Project (MSP), Microsoft Office, LyX

**Languages:** English (professional working proficiency), Turkish — Istanbul dialect (conversational) — Persian (native)

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## EDUCATION

**M.Sc. Mechatronics Engineering** — Amirkabir University of Technology, Tehran, Iran

2024 – Present

**B.Sc. Mechanical Engineering** — Sahand University of Technology, Tabriz, Iran

2018 – 2022